MATAIRANGI CABLE CAR COMPANY LTD

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INTERNAL MEMO: TEMPORARY CAR CAPACITY LIMIT

Ref. MCC-ENG-26/14 · Decision issued 30 September 2026 · Engineering and Operations

TO: Duty supervisors, conductors and drive-house staff

FROM: Fiona Draycott, Chief Engineer

SUBJECT: Haul rope wear on the Mount Victoria line and reduced passenger limit

1. BACKGROUND AND DECISION

At the monthly magnetic and calliper inspection on 28 September 2026, the 24 mm haul rope (installed March 2019) measured 23.2 mm at its narrowest point, 140 m below the Matairangi Lookout station on the steepest (18.0 %) section. The worn rope no longer supports the rated 100 passengers per car at our minimum factor of safety of 7.5. **Decision:** from the first departure on Friday 2 October 2026, each car carries at most **85 passengers** until the new rope is fitted.

2. METHOD: TENSION ON THE STEEPEST SECTION, WITH A DYNAMIC FACTOR FOR AN EMERGENCY STOP

$$T = \varphi\left(m_c + n m_p\right) g\left(\sin\theta + \mu\cos\theta\right) \tag{1}$$

where T is the design rope tension (N), φ = 1.25 the dynamic factor, m_c = 13,500 kg the empty car mass with conductor, n the number of passengers, m_p = 80 kg the standard passenger mass (75 kg before the 2025 review), g = 9.81 m/s², θ = 10.20° for the 18.0 % gradient and μ = 0.012 the rolling resistance.

$$B_e = B_0\left(1 - 2.5\frac{d_0 - d}{d_0}\right) \tag{2}$$

where B_e is the wear-adjusted breaking strength (kN), B_0 = 388 kN the certified breaking load of the new rope, d_0 = 24.0 mm the nominal and d the smallest measured diameter (mm).

$$S = \frac{B_e}{T} \geq 7.5 \tag{3}$$

where S is the factor of safety (B_e and T in the same units).

Worked example (reading of 23.2 mm). Equation (2) gives B_e = 388 × (1 − 2.5 × 0.8/24.0) = 355.67 kN, so the largest permitted tension is 355.67/7.5 = 47.42 kN. With φg(sin θ + μ cos θ) = 2.317 N/kg the car may weigh 20,466 kg, leaving 6,966 kg or 87.07 passengers, rounded down to 85. A full car of 100 gives T = 49.82 kN and S = 7.14.

3. WEEKLY READINGS AND ACTIONS

The rope technician, Hemi Tawhiri, measures every Monday before 06:30. A reading of 23.2 mm or more keeps the limit at 85 passengers per car; 23.0 or 23.1 mm lowers it to 80; 22.8 or 22.9 mm lowers it to 75 and the Chief Engineer is told the same day; below 22.8 mm the line closes until the rope is replaced.

> **Replacement.** The new rope is ordered from Kowhai Rope Works, Petone, at NZD 148,600, with installation at NZD 37,250 and crane hire at NZD 9,480. The line will close for installation from Monday 9 to Friday 13 November 2026; a bus runs every 15 minutes.

Approved for immediate issue: **Fiona Draycott**, Chief Engineer

Fictional document created for testing a question-answering system. All names, figures and organisations are invented.